Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R12-E1_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)

Name	SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)
Description	
Analysis	Stress, strain/ simplified loading

Design

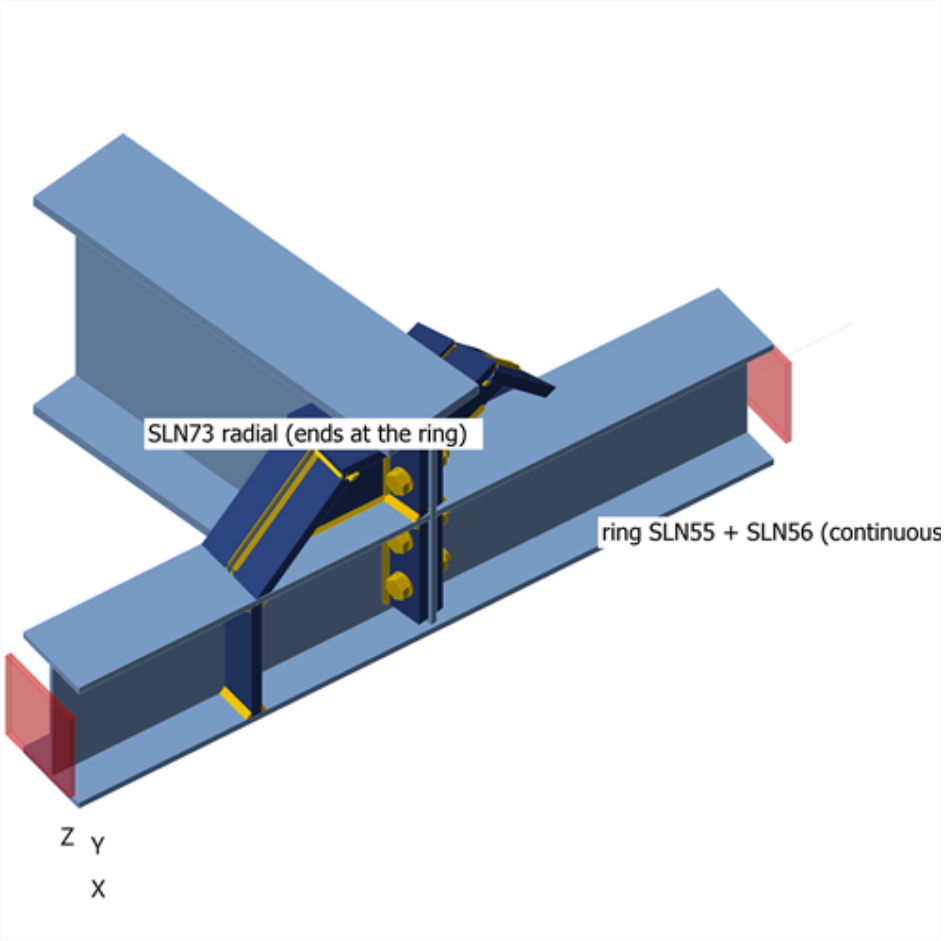
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73 radial (ends at the ring)	1 - UB457x191x98	0.0	0.0	0.0	76	0	0
ring SLN55 + SLN56 (continuous)	2 - UB254x146x43	-100.0	0.0	0.0	0	0	-104

Supports and forces

Name	Support	Forces in	X [mm]
SLN73 radial (ends at the ring) / begin		Position	0
ring SLN55 + SLN56 (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
ring SLN55 + SLN56 (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0



SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous), Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M24 10.9	24	900.0	1000.0	452

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_102	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_103	SLN73 radial (ends at the ring) / Begin	9.6	-0.1	-86.2	-0.1	-45.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_104	SLN73 radial (ends at the ring) / Begin	7.9	0.0	-86.4	-0.1	-45.4	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_105	SLN73 radial (ends at the ring) / Begin	8.8	-0.1	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_106	SLN73 radial (ends at the ring) / Begin	8.0	0.0	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_107	SLN73 radial (ends at the ring) / Begin	1.8	-0.1	-7.8	0.0	-4.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_108	SLN73 radial (ends at the ring) / Begin	1.7	0.1	-8.4	0.0	-4.6	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_109	SLN73 radial (ends at the ring) / Begin	1.4	0.1	-8.1	0.0	-4.5	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_110	SLN73 radial (ends at the ring) / Begin	0.6	-0.1	-8.1	0.0	-4.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_111	SLN73 radial (ends at the ring) / Begin	7.4	-0.1	-61.7	0.0	-32.6	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_112	SLN73 radial (ends at the ring) / Begin	6.3	0.1	-62.3	0.0	-32.8	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_113	SLN73 radial (ends at the ring) / Begin	6.7	0.0	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_114	SLN73 radial (ends at the ring) / Begin	5.6	-0.1	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_115	SLN73 radial (ends at the ring) / Begin	2.4	-0.1	-13.1	0.0	-7.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_116	SLN73 radial (ends at the ring) / Begin	2.1	0.1	-13.6	0.0	-7.4	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_117	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-13.3	0.0	-7.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_118	SLN73 radial (ends at the ring) / Begin	1.1	-0.1	-13.3	0.0	-7.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_119	SLN73 radial (ends at the ring) / Begin	11.2	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_120	SLN73 radial (ends at the ring) / Begin	8.1	0.0	-89.8	-0.1	-47.2	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_121	SLN73 radial (ends at the ring) / Begin	8.5	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_122	SLN73 radial (ends at the ring) / Begin	7.6	0.0	-89.8	-0.1	-47.2	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_123	SLN73 radial (ends at the ring) / Begin	10.8	-0.1	-68.9	0.0	-36.3	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_124	SLN73 radial (ends at the ring) / Begin	6.7	0.1	-68.9	0.0	-36.3	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_125	SLN73 radial (ends at the ring) / Begin	6.2	0.0	-68.9	0.0	-36.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_126	SLN73 radial (ends at the ring) / Begin	4.8	-0.1	-68.9	0.0	-36.3	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_127	SLN73 radial (ends at the ring) / Begin	5.7	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_128	SLN73 radial (ends at the ring) / Begin	2.5	0.1	-20.2	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_129	SLN73 radial (ends at the ring) / Begin	1.5	0.0	-20.3	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_130	SLN73 radial (ends at the ring) / Begin	0.3	-0.1	-20.3	0.0	-10.9	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_135	SLN73 radial (ends at the ring) / Begin	7.7	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_141	SLN73 radial (ends at the ring) / Begin	2.3	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_142	SLN73 radial (ends at the ring) / Begin	1.8	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_143	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_144	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_145	SLN73 radial (ends at the ring) / Begin	10.1	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_146	SLN73 radial (ends at the ring) / Begin	8.3	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_147	SLN73 radial (ends at the ring) / Begin	9.4	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_148	SLN73 radial (ends at the ring) / Begin	9.0	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_151	SLN73 radial (ends at the ring) / Begin	6.9	-0.1	-46.4	0.0	-22.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_152	SLN73 radial (ends at the ring) / Begin	9.3	0.0	-63.5	0.0	-35.9	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_153	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.4	0.0	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_154	SLN73 radial (ends at the ring) / Begin	7.0	-0.1	-46.8	0.0	-22.5	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	6.6 > 5.0%	**Not OK!**
Bolts	60.7 < 100%	OK
Welds	109.1 > 100%	**Not OK!**
Buckling	Not calculated	

Plates

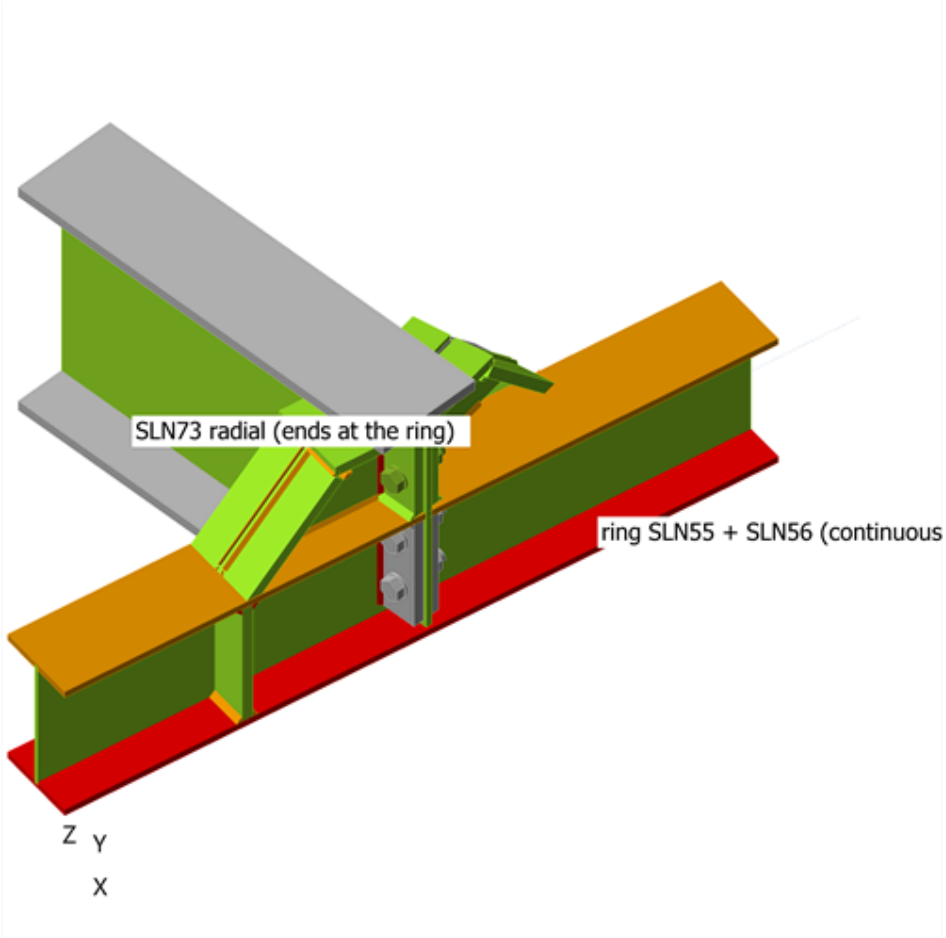
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73 radial (ends at the ring)-bfl 1	19.6	C LC_119	16.9	0.0	0.0	OK
SLN73 radial (ends at the ring)-tfl 1	19.6	C LC_122	180.8	0.0	20.1	OK
SLN73 radial (ends at the ring)-w 1	11.4	C LC_122	275.1	0.0	69.0	OK
ring SLN55 + SLN56 (continuous)-bfl 1	12.7	C LC_153	288.8	6.6	0.0	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	12.7	C LC_119	285.3	4.9	0.0	OK
ring SLN55 + SLN56 (continuous)-w 1	7.2	C LC_121	281.0	2.8	0.0	OK
395x145x20 fin plate upper (SLN55 side)	20.0	C LC_122	275.6	0.3	757.7	OK
395x145x20 fin plate lower outer (SLN55 side)	20.0	C LC_120	157.0	0.0	12.2	OK
395x145x20 fin plate lower inner (SLN55 side)	20.0	C LC_119	216.2	0.0	38.6	OK
395x145x20 fin plate upper (SLN56 side)	20.0	C LC_135	275.2	0.1	803.7	OK
395x145x20 fin plate lower outer (SLN56 side)	20.0	C LC_122	178.3	0.0	17.6	OK
395x145x20 fin plate lower inner (SLN56 side)	20.0	C LC_120	154.3	0.0	43.0	OK
20 ring stiffener at 315 (SLN55 side)	20.0	C LC_122	280.8	2.7	0.0	OK
10 bracket web plate (SLN55 side)	10.0	C LC_122	275.2	0.1	0.0	OK
20 bracket top plate a (SLN55 side)	20.0	C LC_122	275.0	0.0	84.5	OK
20 inclined stiffener a (SLN55 side)	20.0	C LC_122	184.2	0.0	0.0	OK
20 bracket top plate b (SLN55 side)	20.0	C LC_122	275.4	0.2	73.7	OK
20 inclined stiffener b (SLN55 side)	20.0	C LC_122	277.8	1.3	0.0	OK
20 ring stiffener at 315 (SLN56 side)	20.0	C LC_122	276.1	0.5	0.0	OK
10 bracket web plate (SLN56 side)	10.0	C LC_119	275.9	0.4	0.0	OK
20 bracket top plate a (SLN56 side)	20.0	C LC_135	277.4	1.1	0.0	OK
20 inclined stiffener a (SLN56 side)	20.0	C LC_153	275.9	0.4	0.0	OK
20 bracket top plate b (SLN56 side)	20.0	C LC_119	275.1	0.0	255.2	OK
20 inclined stiffener b (SLN56 side)	20.0	C LC_153	275.8	0.4	0.0	OK

Design data

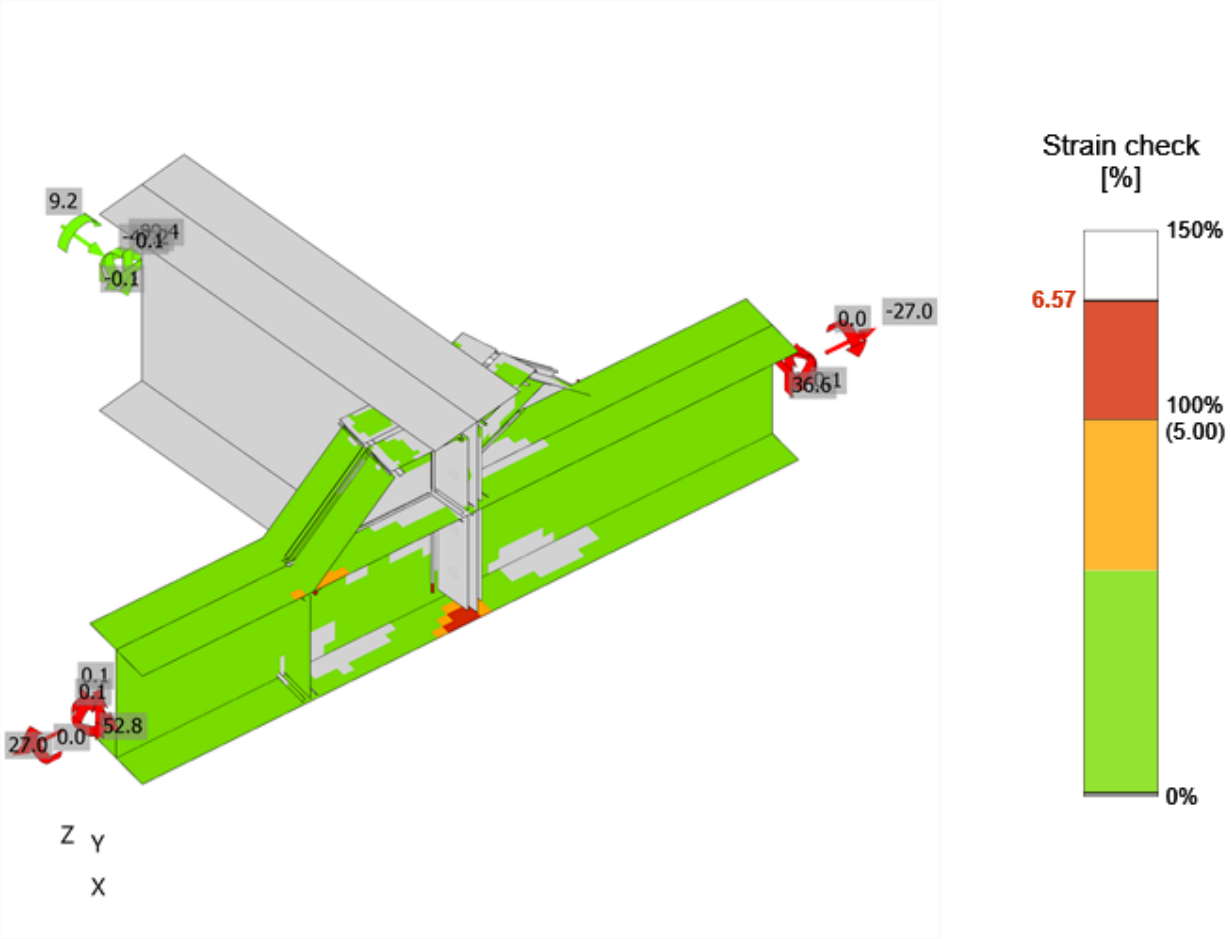
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

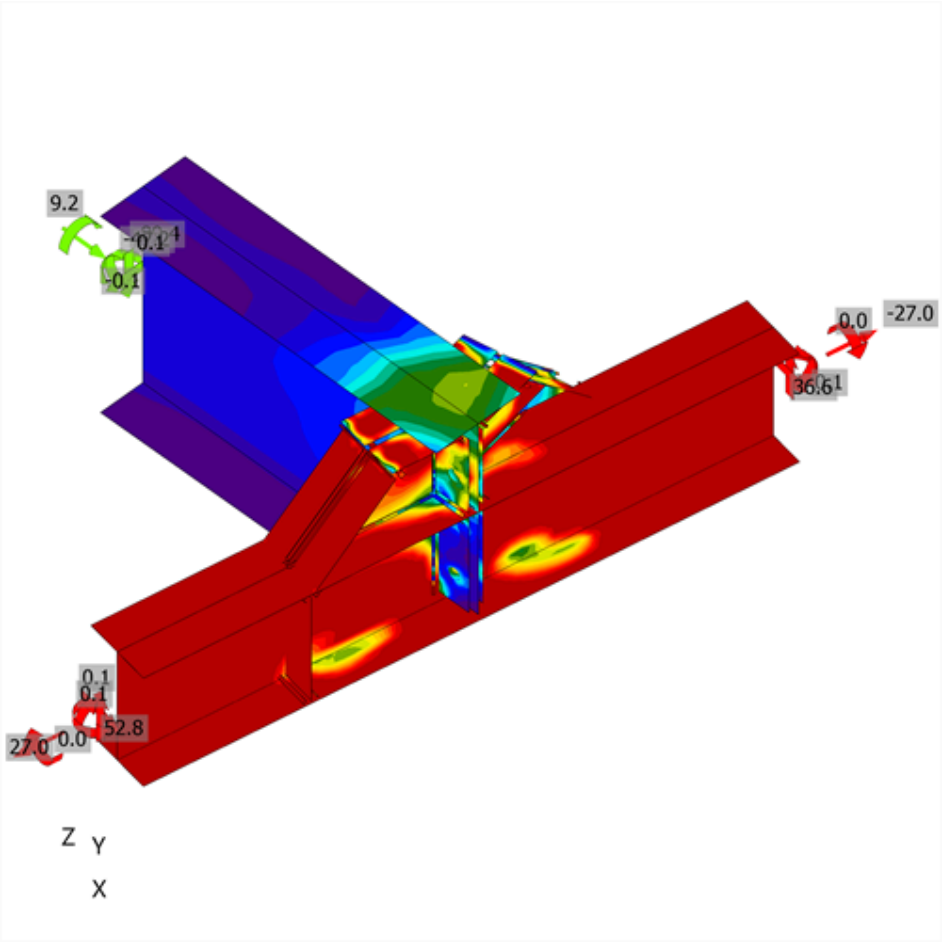
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

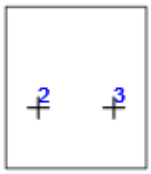
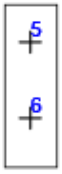
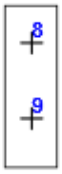


Strain check, C LC_153



Equivalent stress, C LC_153

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M24 10.9 - 1	C LC_122	24.1	38.0	176.5	9.5	39.8	33.7	OK	OK
	B3	M24 10.9 - 1	C LC_122	2.6	38.3	119.3	1.0	60.7	27.9	OK	OK
	B5	M24 10.9 - 1	C LC_122	6.9	10.6	235.3	2.7	9.0	9.4	OK	OK
	B6	M24 10.9 - 1	C LC_153	0.0	35.2	235.3	0.0	29.4	24.9	OK	OK
	B8	M24 10.9 - 1	C LC_119	0.7	13.5	235.3	0.3	11.0	9.7	OK	OK
	B9	M24 10.9 - 1	C LC_153	13.4	33.6	235.3	5.3	28.3	27.5	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M24 10.9 - 1	254.2	492.7	141.2

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN55 side)	20 bracket top plate a (SLN55 side)	-	128	-	-	-	-	-	-	-	-	OK	OK
		-	128	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN55 side)	20 bracket top plate b (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate a (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate b (SLN56 side)	-	128	-	-	-	-	-	-	-	-	OK	OK
		-	128	-	-	-	-	-	-	-	-	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	212	C LC_122	435.3	23.9	-75.7	-161.2	-187.8	107.6	100.0	OK	**Not OK!**
		◢ 4.2 ◣	213	C LC_121	434.3	23.3	-210.5	159.6	150.5	107.3	100.0	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_122	102.1	0.0	18.5	11.5	-56.8	25.2	18.7	OK	OK
		◢ 7.0 ◣	69	C LC_122	145.3	0.0	-67.3	-3.2	-74.3	35.9	25.3	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	213	C LC_122	418.4	13.5	-179.0	205.7	73.3	103.4	100.0	OK	**Not OK!**
		◢ 4.2 ◣	213	C LC_121	429.6	20.4	-208.4	-212.7	-42.3	106.1	100.0	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN56 side)	◢ 7.0 ◣	144	C LC_119	222.7	0.0	-71.7	-92.0	79.7	55.0	36.4	OK	OK
		◢ 7.0 ◣	144	C LC_119	333.0	0.0	210.8	-134.7	-63.3	82.3	55.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_119	145.8	0.0	89.5	63.8	18.4	36.0	22.8	OK	OK
		◢ 7.0 ◣	69	C LC_119	125.1	0.0	38.7	-64.5	-23.7	30.9	21.4	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	213	C LC_122	423.0	16.3	-230.0	-180.8	-96.6	104.5	100.0	OK	**Not OK!**

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
		◢ 4.2 ◣	212	C LC_122	413.3	10.3	-147.8	194.2	109.3	102.1	100.0	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_119	149.3	0.0	-116.5	-52.3	12.9	37.6	25.7	OK	OK
		◢ 7.0 ◣	69	C LC_119	106.1	0.0	4.9	59.3	-15.3	26.2	18.5	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	213	C LC_122	417.8	13.1	182.5	193.6	-98.0	103.2	100.0	OK	**Not OK!**
		◢ 4.2 ◣	212	C LC_122	418.8	13.7	186.2	-181.1	118.8	103.5	100.0	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	146	C LC_122	396.7	0.0	129.2	84.6	199.3	98.0	82.3	OK	OK
		◢ 7.0 ◣	146	C LC_122	339.5	0.0	-81.6	34.0	187.2	83.9	60.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	146	C LC_119	397.1	0.3	91.9	-222.7	12.1	98.1	87.4	OK	OK
		◢ 7.0 ◣	146	C LC_119	410.7	8.7	-203.9	200.2	-47.7	101.5	100.0	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	288	C LC_153	397.9	0.8	72.2	-99.4	202.9	98.3	72.1	OK	OK
		◢ 4.2 ◣	287	C LC_153	397.6	0.6	-143.6	-77.1	199.7	98.2	65.6	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_119	404.7	5.0	-151.2	192.1	-100.3	100.0	98.4	OK	**Not OK!**
		◢ 7.0 ◣	67	C LC_153	404.2	4.7	-133.7	-204.8	81.0	99.9	90.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_119	405.4	5.4	165.2	-192.1	-93.6	100.2	99.6	OK	**Not OK!**
		◢ 7.0 ◣	67	C LC_153	403.1	4.0	140.0	201.3	84.3	99.6	92.2	OK	OK
395x145x20 fin plate upper (SLN55 side)	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	167	C LC_122	396.8	0.1	-228.3	-186.8	-15.4	98.1	74.9	OK	OK
		◢ 4.2 ◣	167	C LC_135	357.9	0.0	-40.6	186.8	85.2	88.4	60.1	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
395x145x20 fin plate upper (SLN56 side)	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	167	C LC_153	405.6	5.6	157.7	-193.6	-95.3	100.2	90.8	OK	**Not OK!**
		◢ 4.2 ◣	167	C LC_119	402.0	3.3	-302.5	-120.7	-93.7	99.3	84.5	OK	OK
20 bracket top plate a (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_122	192.5	0.0	-189.6	9.7	-16.3	61.3	31.0	OK	OK
		◢ 7.0 ◣	67	C LC_122	220.2	0.0	153.1	-88.7	-22.0	54.4	37.7	OK	OK
20 bracket top plate b (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_122	204.2	0.0	-170.7	-60.6	-22.8	55.1	42.3	OK	OK
		◢ 7.0 ◣	67	C LC_122	272.5	0.0	-190.9	112.1	-4.0	67.3	49.5	OK	OK
20 bracket top plate a (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.6	0.6	16.8	-44.8	-224.9	98.2	82.9	OK	OK
		◢ 7.0 ◣	67	C LC_135	396.8	0.1	-79.7	48.8	-219.0	98.0	70.8	OK	OK
20 bracket top plate b (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_122	398.1	0.9	-45.9	7.1	-228.2	98.4	88.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.8	0.1	-40.7	-4.4	-227.8	98.0	71.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN55 side)	◢ 7.0 ◣	141	C LC_122	357.9	0.0	-177.4	-103.2	-146.9	88.4	53.2	OK	OK
		◢ 7.0 ◣	141	C LC_122	230.1	0.0	-34.4	-65.5	113.9	56.9	33.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	65	C LC_122	145.4	0.0	-20.1	-33.0	76.3	35.9	22.3	OK	OK
		◢ 7.0 ◣	65	C LC_122	134.7	0.0	101.8	-27.5	42.8	33.3	23.7	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	37	C LC_121	420.1	14.5	-129.3	-216.5	-79.8	103.8	101.8	OK	**Not OK!**
		◢ 7.0 ◣	37	C LC_153	410.8	8.8	-298.8	158.9	-35.2	101.5	100.2	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	37	C LC_121	441.6	27.8	-108.2	-228.6	-94.0	109.1	103.7	OK	**Not OK!**
		◢ 7.0 ◣	37	C LC_120	439.9	26.7	-303.8	179.2	-40.4	108.7	103.7	OK	**Not OK!**

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	189	C LC_122	312.7	0.0	-301.5	-38.6	-28.3	97.4	57.0	OK	OK
		◢ 4.2 ◣	189	C LC_122	397.1	0.3	-102.2	213.2	-60.1	98.1	72.5	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	39	C LC_122	434.6	23.4	291.7	-178.2	53.1	107.4	104.3	OK	**Not OK!**
		◢ 7.0 ◣	39	C LC_122	404.3	4.8	163.7	213.2	-9.8	99.9	94.9	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 4.2 ◣	222	C LC_122	299.8	0.0	-69.3	43.4	162.7	74.1	48.7	OK	OK
		◢ 4.2 ◣	222	C LC_122	247.9	0.0	125.6	-12.9	122.7	61.3	38.1	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 4.2 ◣	222	C LC_122	316.2	0.0	-261.1	-72.0	-73.7	84.3	38.5	OK	OK
		◢ 4.2 ◣	222	C LC_122	379.4	0.0	167.5	-49.7	190.2	93.8	57.2	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.4	1.1	135.2	107.8	187.6	98.4	93.4	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.0	0.9	150.9	-184.8	-105.3	98.3	74.5	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.1	0.9	-273.3	-130.8	104.0	98.4	91.7	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.3	1.1	-72.3	195.7	-113.4	98.4	78.3	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate a (SLN55 side)	◢ 4.2 ◣	65	C LC_122	396.7	0.1	-168.4	-207.4	2.3	98.0	71.9	OK	OK
		◢ 4.2 ◣	65	C LC_122	396.8	0.1	-215.1	-183.8	-57.2	98.0	67.1	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate b (SLN55 side)	◢ 4.2 ◣	68	C LC_122	398.3	1.0	-183.8	203.6	12.9	98.4	89.1	OK	OK
		◢ 4.2 ◣	68	C LC_122	398.1	0.9	41.7	224.9	-41.0	98.4	81.1	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate a (SLN56 side)	◢ 4.2 ◣	68	C LC_135	398.8	1.4	-263.6	172.6	8.6	98.6	90.3	OK	OK
		◢ 4.2 ◣	68	C LC_135	398.5	1.2	-209.4	192.7	34.4	98.5	90.4	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate b (SLN56 side)	◢ 4.2 ◣	65	C LC_122	397.3	0.4	-153.7	-206.3	46.4	98.2	81.2	OK	OK
		◢ 4.2 ◣	65	C LC_122	397.4	0.5	203.7	-177.9	84.6	98.2	85.9	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_C	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3

Item	Value	Unit	Reference
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5